SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE CODE: CSA1509

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Dave's Taxonomy: Imitation (Level 1)

Question Count: 5 **Total Marks: 25**

Code of Conduct:

I, **ASHIRA BRIDGET SUSAN S / 192524316,** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks
Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of data centers in delivering cloud services.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

ANSWERS

1. Cloud Computing Definition and Characteristics:

Cloud Computing is the delivery of computing services—including servers, storage, databases, networking, software, and analytics—over the internet ("the cloud"). Rather than owning and maintaining physical data centers and servers, organizations rent access to these resources from a cloud provider, allowing for faster innovation, flexible resources, and economies of scale.

Four key characteristics that differentiate cloud computing from traditional on-premise computing include:

On-Demand Self-Service:

- *Cloud:* Users can instantly provision computing capabilities (like server time and network storage) automatically through a web interface without requiring human interaction from the service provider.
- *On-Premise:* Acquiring new resources requires a lengthy procurement process involving hardware purchasing, shipping, manual installation, and IT configuration.

Rapid Elasticity (Scalability):

- *Cloud:* Resources can be scaled outward or inward automatically to handle sudden spikes or drops in demand. To the consumer, the available resources often appear unlimited and can be appropriated in any quantity at any time.
- *On-Premise:* Hardware limits are fixed. Scaling up requires buying new physical servers (which may sit idle during off-peak times), and scaling down is impossible without wasting the capital already spent.

Resource Pooling:

- *Cloud:* The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model. Physical and virtual resources are dynamically assigned and reassigned according to consumer demand, abstracting the exact physical location of the data.
- *On-Premise:* Resources are dedicated to a single organization (single-tenant). Servers are often underutilized, sitting at 10-15% capacity, creating massive inefficiencies.

Measured Service (Pay-as-you-go):

- *Cloud:* Cloud systems automatically control and optimize resource use by leveraging a metering capability. Resource usage can be monitored, controlled, and reported, providing transparency for both the provider and the consumer. You only pay for what you consume (Operating Expense or OpEx).
- *On-Premise:* Requires massive upfront capital expenditure (CapEx) to purchase hardware and software licenses, regardless of how much the hardware is actually used.

2. Evolution of Cloud Computing:

Modern cloud platforms did not emerge overnight; they are the culmination of parallel evolutions in hardware architectures and internet-based software methodologies.

Phase 1: Hardware Evolution

- **Mainframes to Minicomputers (1950s–1970s):** Computing began with massive, expensive mainframes accessed via "dumb terminals." This established the very first concept of shared computing power.
- **Personal Computers & Client-Server Architecture (1980s–1990s):** Processing power moved to the user's desk. However, this created decentralized data silos and management nightmares, pushing the industry to connect these PCs to central servers.
- **Grid and Cluster Computing (1990s–2000s):** To solve complex problems, engineers started linking multiple separate computers together to act as a single powerful machine (clustering) or distributing tasks across geographically dispersed computers (grid computing). This laid the foundation for distributed cloud architecture.

Phase 2: Internet and Software Evolution

- **Web 1.0 to Web 2.0 (1990s–2000s):** The internet shifted from serving static HTML pages (Web 1.0) to dynamic, interactive applications running in the browser (Web 2.0). This proved that complex software did not need to be installed locally.
- **Service-Oriented Architecture (SOA):** Software design shifted to creating modular applications where components provide services to other components via a communications protocol, usually over a network. This decoupled software from underlying hardware.
- **Virtualization Revolution (2000s):** The perfection of x86 virtualization allowed multiple operating systems to run on a single physical server, maximizing hardware efficiency.

How these enabled modern platforms:

Hardware evolution provided the raw, dense compute power and cheap storage necessary to build massive data centers. Simultaneously, the software evolution provided high-speed internet protocols, browser-based interfaces, and virtualization software. By combining grid computing's distributed nature with virtualization and web-based delivery, tech giants (like Amazon and Google) realized they could rent out their excess data center capacity as a "utility," creating modern cloud computing.

3. Server Virtualization and Cloud Computing:

The Role of Server Virtualization:

Server virtualization is the foundational technology that makes modern cloud computing possible. It involves using software—called a Hypervisor—to abstract and partition the physical hardware of a single server into multiple, isolated Virtual Machines (VMs).

- **Hardware Abstraction:** It hides the physical hardware details from the operating system, allowing a VM to be easily moved from one physical server to another without downtime.
- **Resource Maximization:** Instead of running one application on one physical server (which typically utilizes only 15% of the CPU), virtualization allows dozens of VMs to share the same CPU, RAM, and storage, driving utilization up to 80% or more.
- **Isolation and Security:** Each VM acts as an entirely separate machine. If one VM crashes or is compromised, the other VMs on the same physical server remain unaffected.

Virtualization vs. Cloud Computing:

While highly related, virtualization and cloud computing are distinct concepts. Virtualization is the engine, while cloud computing is the vehicle.

Feature	Virtualization (The Enabling Technology)	Cloud Computing (The Service Model)
Definition	Technology that separates compute environments from physical infrastructure.	A service delivery model that provides computing resources over the internet.
Primary Goal	To maximize the utilization of physical hardware.	To deliver IT resources as a metered, self-service utility.
Scope	Operates strictly at the infrastructure level (creating VMs).	Operates at the orchestration layer, adding billing, self-service portals, and APIs.
Dependency	Can exist without cloud computing (e.g., a private on-premise virtualized server).	Cannot exist without virtualization; it is built entirely on top of virtualized resources.

4. The Role of Data Centers in Cloud Services:

Data centers are the physical manifestations of the cloud. When a user provisions a resource "in the cloud," it is physically spinning up on a server located within a massive, highly secure data center. Their roles in delivering cloud services include:

- **Hosting Compute and Storage Infrastructure:** Data centers house tens of thousands of tightly packed server racks, massive SANs (Storage Area Networks), and enterprise-grade networking equipment. This hardware executes the actual processing and stores the data for cloud consumers.

- **Ensuring High Availability and Redundancy:** Cloud providers design data centers with N+1 or 2N redundancy for every critical component. They feature industrial-grade Uninterruptible Power Supplies (UPS), massive diesel backup generators, and advanced HVAC cooling systems to ensure servers stay online 24/7/365, even during regional grid failures.

- **Providing Extreme Security:** Data centers act as digital fortresses. They provide physical security measures far beyond what a normal business could afford, including biometric scanners, vehicle traps, 24/7 armed security, and inert gas fire suppression systems.

- **Enabling Global Reach and Low Latency (Availability Zones):** Cloud providers build data centers in strategic regions worldwide. By interconnecting these centers via high-speed, dedicated fiber optic cables, they allow businesses to host applications physically closer to their end-users, reducing latency and ensuring compliance with local data sovereignty laws.

- **Massive Scalability:** The sheer physical size of these facilities allows providers to constantly add new racks of servers. This creates the "illusion of infinite capacity" that cloud users rely on when they rapidly scale their applications.

5. Differentiating IaaS, PaaS, SaaS, and XaaS:

Cloud computing services are categorized into different models based on how much of the IT stack the cloud provider manages versus how much the user manages.

1. IaaS (Infrastructure as a Service):

- **Concept:** The most fundamental cloud model. The provider supplies the raw, virtualized computing resources (servers, storage, and networking). The user is responsible for installing the operating system, middleware, databases, and applications.
- **Best for:** System administrators who need complete control over the environment and OS without buying physical hardware.
- **Real-World Example:** Amazon EC2 (Elastic Compute Cloud). A developer rents a blank virtual server, chooses to install Ubuntu Linux on it, and sets up their own web server from scratch.

2. PaaS (Platform as a Service):

- **Concept:** The provider manages the underlying infrastructure, operating systems, and runtime environments. The user only manages the application code and the data. It provides a ready-made framework for software development.
- **Best for:** Developers who want to focus strictly on writing code without worrying about server maintenance, patching, or load balancing.
- **Real-World Example**: Google App Engine or Heroku. A developer uploads their Python or Node.js code, and the platform automatically handles the server provisioning, scaling, and execution.

3. SaaS (Software as a Service):

- **Concept:** The cloud provider manages the entire technology stack, from infrastructure to the application itself. The software is delivered as a fully functional, ready-to-use product accessible directly via a web browser.
- **Best for:** End-users and businesses that need a specific software tool immediately without any installation or maintenance requirements.
- **Real-World Example:** Google Workspace (Gmail, Google Docs) or Salesforce. You simply log in via a browser to use the email or CRM tools; you do not manage the servers or the software updates.

4. XaaS (Anything/Everything as a Service):

- **Concept:** A blanket acronym recognizing the expansion of the cloud model beyond the standard three (IaaS, PaaS, SaaS). It refers to the delivery of almost any IT function, business process, or technology over the internet on a subscription basis.
- **Best for:** Organizations looking to shift specific, complex IT responsibilities to specialized third-party providers.
- **Real-World Example**: DBaaS (Database as a Service) like Amazon RDS, where the provider manages database backups and patching; or DRaaS (Disaster Recovery as a Service), where a provider automatically replicates a company's infrastructure to a secondary location in case of a disaster.

Short Answer Test Rubric:

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow